

Retail Customer Analysis Report

Summary of Findings

This project analyzed a retail customer dataset to understand customer characteristics, purchasing behavior, and factors influencing customer value. The dataset was first cleaned by removing duplicate records and handling missing values in the **Income** column using **median imputation**. The median was selected because it is a robust measure of central tendency and is less affected by extreme values, making it suitable for income data. Additionally, a new variable, **Customer Lifetime Value (CLV)**, was created using the formula:

$$\text{CLV} = \text{Purchases} \times \text{Member_Years} \times 15$$

to estimate each customer's potential long-term value to the business.

Correlation analysis was performed to examine relationships between key variables. The correlation between **Income** and **Satisfaction** was **-0.013**, indicating virtually **no linear relationship**. This suggests that customer income does not significantly influence satisfaction levels. In contrast, the correlation between **Member_Years** and **CLV** was **0.756**, indicating a **strong positive relationship**. This finding suggests that customers who remain with the company for longer generally generate greater lifetime value.

An independent-samples **t-test** compared the average income of customers with **Satisfaction scores of 7 or higher** and those with **scores below 7**. The analysis produced a **t-statistic of -0.213** and a **p-value of 0.8315**. Since the p-value is greater than the significance level of **0.05**, the difference is **not statistically significant**. This indicates that there is no sufficient evidence to conclude that customer income differs between the two satisfaction groups.

Visualization Summary

Two visualizations were developed to support the analysis. The **bar chart** showed the average income across different regions, highlighting regional variations in customer income. The **scatter plot** illustrated the relationship between **Age** and **Purchases**. The data points were widely dispersed with no clear trend, suggesting that age is not a strong predictor of purchasing behavior. Customers across different age groups made varying numbers of purchases.

Recommendations

1. **Strengthen customer retention initiatives:** Since longer membership is strongly associated with higher Customer Lifetime Value, the company should invest in loyalty programs, personalized rewards, and retention campaigns to encourage long-term customer relationships.
2. **Improve customer experience rather than targeting customers by income:** As income showed virtually no relationship with customer satisfaction, the company should focus on improving product quality, customer service, and personalized engagement to increase customer satisfaction and encourage repeat purchases.

Conclusion

The analysis indicates that customer loyalty plays a more important role in increasing customer value than customer income. While income does not significantly influence satisfaction, retaining customers over a longer period can substantially improve their lifetime value. By focusing on customer retention and enhancing the overall customer experience, the company can build stronger customer relationships and achieve sustainable business growth.